Jiacheng Zhang



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EDUCATION

The University of Manchester **Sep 2026 – Sep 2027**

MSc Robotics | Expected graduation September 2027

University of Shanghai for Science and Technology **Sep 2022 – Jul 2026**

BEng Robotics Engineering | Overall average 88.03/100

TECHNICAL SKILLS

Programming and tools: C++, C, Python, OpenCV, ROS, Arduino, MATLAB, Simulink, Siemens NX (UG), SolidWorks

Vision and robotics: Camera and hand–eye calibration, coordinate transformations, HSV K-means clustering, Canny and Hough detection, sensor integration

SELECTED TECHNICAL PROJECTS

All Terrain Collaborative Robots **2025**

- Built hardware and integrated grayscale, near-infrared, colour and tactile sensors; tuned operating parameters and modified state-machine logic within an existing framework during later-stage testing.
- Team awarded national second prize and Shanghai first prize in the China Robot and Artificial Intelligence Competition's All-Terrain Collaborative Robots event.

Vision Based Localisation for Robotic Assembly **Sep – Dec 2024**

- Built on a supervisor-provided project to develop a C++/OpenCV rivet-hole detection pipeline using preprocessing, Canny edges and Hough circle detection.
- Used Zhang's calibration method and ArUco markers for camera calibration and extrinsic estimation; implemented coordinate transformations for spatial localisation.

Vision Guided Six Axis Robotic Arm for Tetris **2024**

- Developed a C++/RealSense pipeline with HSV K-means clustering; performed hand–eye calibration and transmitted transformed 3D target coordinates via ROS to a SixArm robotic arm.
- Team awarded second prize in the China University Intelligent Robot Innovation Competition and first prize in the preliminary Tetris event.

INDUSTRY EXPERIENCE

Tenneco | Finance and Logistics Intern **Jul – Aug 2025**

- Conducted time studies of material handling, AGV delivery, inventory and final assembly; supported standard work analysis for the P3 lean manufacturing improvement project.

Zhejiang Shenxin Packaging | Mould Workshop Intern **Jul – Sep 2024**

- Assisted with CAD/CAM mould design, CNC toolpath preparation, assembly, debugging, dimensional inspection and functional testing.

Zhejiang Shenxin Packaging | Intelligent Manufacturing Intern **Jul – Sep 2023**

- Supported image-based training, parameter tuning and testing of a neural-network system detecting plastic-bottle damage through bright-spot features.

ADDITIONAL ACHIEVEMENTS AND LANGUAGES

RAICOM Football: national second prize and Shanghai first prize, 2024. Academic Excellence Scholarships: 2022–2025.

Mandarin: native | English: IELTS 7.0 overall, November 2024